

3. (a) The diagrams represent five substances, **A-E**.

**A****B****C****D****E**

Key carbon hydrogen oxygen

(i) State which diagram, **A-E**, represents an element. Give a reason for your answer. [2]

Element

Reason

(ii) State which diagram, **A-E**, represents carbon dioxide, CO_2 . [1]

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(iii) Give the formula of the substance represented by diagram **C**. [1]

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(b) When copper(II) sulfate solution reacts with sodium hydroxide, it forms copper(II) hydroxide, $\text{Cu}(\text{OH})_2$, and sodium sulfate, Na_2SO_4 .

(i) Write a **word** equation for the reaction taking place. [1]

(ii) Give the number of atoms of sulfur found in the formula Na_2SO_4 . [1]

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(iii) Give the **total** number of atoms in the formula $\text{Cu}(\text{OH})_2$. [1]

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(c) Iron(III) sulfate reacts with sodium hydroxide to produce iron(III) hydroxide.

Write the formula of iron(III) hydroxide. [1]

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